

Table of Contents

Chapter 1: Introduction	
➤ Overview of the project	
➤ Background of Data Communication	
➤ Problem statement	
➤ Objectives	
➤ Scope of the project	
Chapter 2: Literature Review / Existing System	
➤ Description of existing network/system	
➤ Manual or old networking issues	
➤ Limitations of current system	
Chapter 3: Proposed System / Network Design	
➤ Proposed network solution	
➤ Network architecture	
➤ Network topology used (Star, Bus, Mesh, etc.)	
➤ Advantages of proposed system	
Chapter 4: Network Components & Technologies	
➤ Network devices (Router, Switch, Hub, Modem)	
➤ Transmission media (UTP, Fiber, Wireless)	
➤ Network types (LAN, MAN, WAN)	

Chapter 5: Protocols & IP Addressing

- OSI Model (layer-wise explanation)
- TCP/IP Model
- Protocols used (TCP, UDP, HTTP, FTP, DNS)
- IP addressing (IPv4 / IPv6)

Chapter 6: Implementation / Simulation

- Tools used (Packet Tracer / GNS3)
- Network configuration steps
- Screenshots of simulation
- Data flow explanation

Chapter 7: Testing & Performance Analysis

- Network testing methods
- Ping & traceroute results
- Throughput, delay, packet loss (basic)

Chapter 8: Conclusion & Future Work

- Project summary
- Advantages
- Limitations
- Future enhancements

Chapter # 1: INTRODUCTION

1- Overview of the Project:

High availability is a key requirement for modern enterprise networks, especially for organizations providing web-based services. Any downtime can lead to service disruption, financial loss, and damage to reputation. This project focuses on building a network that ensures continuous availability of web services even in the presence of failures.

The project demonstrates how dynamic routing, load balancing, and failover work together to create a resilient network. Multiple paths and servers are used so that traffic can be rerouted automatically when a failure occurs.

This project focuses on the design and implementation of a high-availability (HA) web server network, a critical requirement for modern digital enterprises where downtime results in significant financial and reputational loss. In a standard single-server setup, any hardware failure or traffic spike can lead to service interruption. This project addresses these vulnerabilities by creating a redundant architecture using Cisco Packet Tracer. By integrating dynamic routing protocols and load-balancing techniques, the network is designed to automatically reroute traffic and distribute workloads across multiple server nodes. The core objective is to ensure that the web services remain accessible even if a specific path or server fails, providing a seamless experience for the end-user.

2- Background of Data Communication:

Data communication involves the transmission of digital data between devices over communication channels. Early networks were simple and static, relying on manual configurations. However, with the growth of the internet and cloud services, networks have become complex and dynamic.

Modern data communication systems rely on:

- Routing protocols
 - Redundant paths
 - Intelligent traffic distribution
- to ensure reliability, scalability, and performance.

3- Problem Statement:

The primary challenge in traditional network environments is the "Single Point of Failure" (**SPOF**). When a network relies on a single path to reach a server, or a single server to handle all requests, it becomes highly fragile. If a router along that path malfunctions or the server encounters a software crash, the entire system goes offline. Furthermore, as user demand grows, a single server often faces "bottlenecking," where its CPU and RAM are overwhelmed by concurrent requests, leading to slow response times or total denial of service. There is a pressing need for a structured networking solution that utilizes redundant physical links and intelligent traffic management to maintain 99.9% uptime.

Traditional network designs suffer from several issues:

- Single path routing
- Single server dependency
- Manual failure recovery
- High downtime

If a router or link fails, the entire service becomes unavailable. Such designs are unsuitable for mission-critical services.

4- Objectives:

The main goals of this project are:

- To design a **ECMP** multi-path network topology that eliminates single points of failure between the client LAN and the server farm.
- To implement dynamic routing protocols (such as **OSPF**) that allow the network to "heal" itself by calculating alternative routes in real-time if a link goes down.
- To establish a load-balancing mechanism that distributes incoming HTTP/HTTPS traffic across three distinct server nodes to optimize resource utilization.
- To verify the failover capabilities of the system through rigorous simulation, ensuring that the transition between primary and backup paths is transparent to the user.
- To analyse network performance

5- Scope of the Project:

The scope of this project encompasses the end-to-end design, configuration, and validation of a high-availability infrastructure for web services. It specifically focuses on the integration of Layer 3 redundancy and service-level reliability. The project involves setting up a simulated enterprise environment featuring a client-side Local Area Network (LAN), a redundant core transit network (the "Internet" simulation), and a multi-node server farm.

The technical scope includes the implementation of the OSPF (Open Shortest Path First) dynamic routing protocol to manage path selection, the configuration of subnets using Variable Length Subnet Masking (VLSM) to optimize IP address space, and the establishment of redundant physical links to eliminate single points of failure. The project is bounded by the Cisco Packet Tracer simulation environment, focusing on the accessibility of HTTP services across three replicated server instances.

The project does not include: Security policies, Real hardware deployment.

Chapter # 2: LITERATURE REVIEW /

EXISTING SYSTEM

1- Description of Existing Network/System:

Traditional or "existing" network systems often rely on linear or "hub-and-spoke" architecture. In such a system, a single router typically connects the local network to the internet, and a single server hosts the required applications. The path from the client to the server is static; data packets follow a fixed route regardless of the health of intermediate nodes. These systems are usually characterized by simplicity and low initial setup costs, making them common in small-scale environments where high uptime is not considered a critical requirement.

Existing systems often use static routing and centralized servers. Such networks are easy to design but lack flexibility and reliability. Traffic always follows the same path regardless of congestion or failure.

2- Manual or Old Networking Issues:

Older networking models rely heavily on manual intervention and static configurations. One of the primary issues is the lack of "intelligence" in the network. If a cable is unplugged or a router interface fails, the network has no automated way to recover. Administrators must manually reconfigure routing tables or swap hardware, a process that can take minutes or even hours, leading to significant downtime. Additionally, manual IP management in these systems often leads to IP conflicts and difficulty in scaling as the number of devices grows.

Moreover, they face these issues as well:

- Manual route configuration
- No automatic rerouting
- High downtime during failures
- Poor scalability

3- Limitations of Current System:

The limitations of the existing systems are primarily centred around "Single Points of Failure" (SPOF). Key limitations include:

- **Zero Redundancy:** If the primary link fails, the entire service becomes unreachable.
- **Poor Scalability:** A single server cannot handle sudden spikes in web traffic, leading to high latency or system crashes.
- **High Latency:** Without dynamic path selection, traffic may take a suboptimal path, increasing the time it takes for a user to receive data.
- **Maintenance Downtime:** In a non-redundant system, performing hardware upgrades or software patches requires taking the entire service offline, as there is no secondary server to take over the load.

Chapter # 3: PROPOSED SYSTEM /

NETWORK DESIGN

1- Proposed Network Solution:

The proposed system introduces a High-Availability (HA) framework designed to ensure that web services remain operational 24/7. This solution utilizes a "Redundant Core" approach, where multiple routers are interconnected to provide at least two distinct paths to any destination. By implementing dynamic routing, the system can automatically detect failures and reroute traffic in milliseconds. To solve the server bottleneck issue, the proposed system replicates the web service across three different server nodes, ensuring that if one server fails, the others continue to process requests. The proposed network introduces:

- Multiple web servers
- Redundant routing paths
- Dynamic routing using OSPF
- Load balancing using ECMP

This eliminates single points of failure and improves performance.

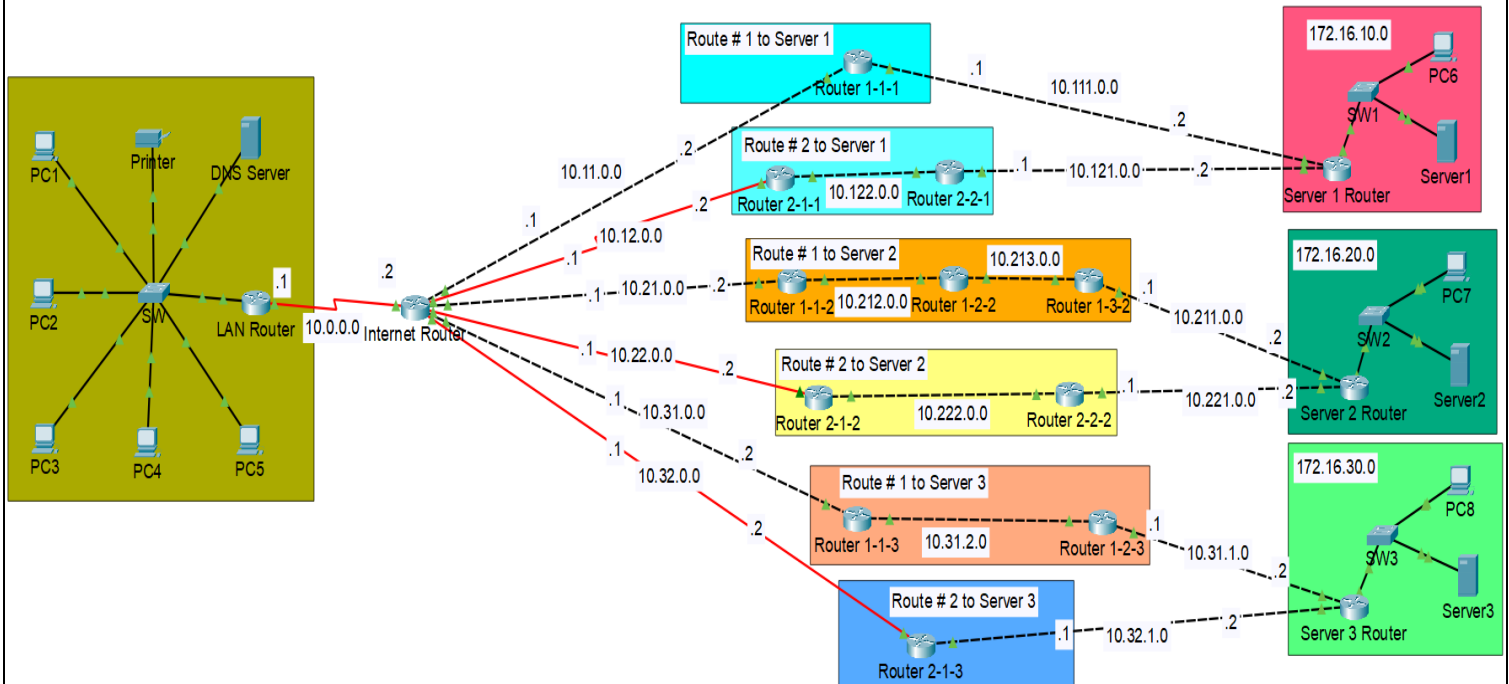
2- Network Architecture:

The architecture follows a three-tier hierarchical model:

1. **Access Tier:** Consists of the user LAN where PCs and local resources reside.
2. **Core/Transit Tier:** A mesh of routers representing the Internet or a private WAN, responsible for high-speed packet switching and path redundancy.

3. **Server Tier:** A dedicated segment containing the server farm, protected by inbound routers that manage traffic distribution from the redundant core links.

Network Topology:



3-Network Topology Used:

Following Topologies are used:

- **Star topology** in LAN, SW is used as the central device and PCs, printer and server are the end devices
- **Partial mesh topology** in WAN, the Internet Router is the Default Gateway of the LAN, it acts as the Route to connect to the Internet and as we know on internet there is no certain topology followed, so, Partial mesh has been used to simulate the network in order to deliver the idea.

This combination provides scalability and redundancy. It ensures that every critical router is connected to at least two others. This provides the redundancy in possible paths necessary for failover operations.

4-Advantages of Proposed System:

- **Maximum Uptime:** The primary advantage is the elimination of downtime. The system "self-heals" using dynamic protocols.
- **Load Distribution:** By spreading traffic across multiple servers, the system prevents any single node from becoming overwhelmed.
- **Optimization:** Using /30 CIDR masks for point-to-point links ensures that IP addresses are used efficiently, while /24 masks in the LAN/Server zones allow for growth.
- **Future-Proofing:** The design is modular, meaning additional servers or routers can be added to the mesh without reconfiguring the entire network.

Chapter # 4: NETWORK COMPONENTS & TECHNOLOGIES

1-Network Devices:

- **Routers:** The "brain" of the network. We utilize Cisco 2911 series routers to perform Layer 3 functions, maintain routing tables, and execute the OSPF algorithm for path selection.
- **Switches:** Cisco 2960 switches are used at the Access and Server tiers to provide high-speed Layer 2 connectivity for end-devices (PCs and Servers) within their respective subnets.
- **Servers:** Three dedicated server nodes are configured to provide HTTP services, acting as the destination for all client requests.
- **End Devices:** PCs, Printers, and DNS servers that represent the user side of the network.

2-Transmission Media:

- **Copper (UTP - Category 6):** Used for all local connections within the LAN and the Server farm. It is also used in one of redundant paths leading to each server network. Cat6 supports 1Gbps speeds, which is ideal for high-speed internal data transfer.
- **Serial Links:** Simulated for long-distance router-to-router communication. Fiber is the standard for modern high-availability cores due to its high bandwidth and immunity to electromagnetic interference.
- **Wireless:** While the core is wired for stability, the architecture can support Wireless Access Points at the Access tier for mobile client connectivity.

3-Network types:

- **Local Area Network (LAN):** The 192.168.10.0 segment represents the internal office network where users access resources. The 172.16.10.0, 172.16.20.0 and the 172.16.30.0 is the segment used for servers.
- **Wide Area Network (WAN):** The 10.x.x.x series of subnets simulate a WAN or Internet backbone, connecting the geographically distant client LAN to the central Server Zone.

Chapter # 5: PROTOCOLS & IP

ADDRESSING

1-OSI Model (layer-wise explanation):

The Open Systems Interconnection (OSI) model is a conceptual framework used to describe the functions of a networking system in seven distinct layers. In the project titled *"High Availability Web Server Network Using Dynamic Routing, Load Balancing, and Failover"*, which implements OSPF as the dynamic routing protocol to ensure redundancy and high availability across multiple paths to three web servers, each OSI layer plays a specific role. The network topology includes a local LAN, an Internet edge router, multiple transit routers with redundant paths, and three remote server sites, with load balancing and failover achieved through OSPF cost manipulation and equal-cost multipath routing.

Below is a layer-by-layer explanation of how data flows (e.g., an HTTP request from a client PC in the local LAN to one of the web servers) and how the project's design maps to each OSI layer.

Physical Layer (Layer 1)

The Physical Layer is responsible for the transmission and reception of raw bit streams over a physical medium.

In the project:

- This layer involves the physical cabling (Ethernet cables), connectors, and network interface cards (NICs) on all devices: PCs, printers, servers, switches (SW1, SW2, SW3), and routers.

- Signals are transmitted as electrical/optical pulses over Ethernet links (e.g., 100 Mbps or 1 Gbps links assumed in the topology).
- No intelligence is added here; it simply provides the medium for connectivity between directly connected devices (e.g., PC1 to SW1, SW1 to LAN Router, or inter-router serial/Ethernet links).
- High availability at this layer is implicitly supported by using multiple physical links for redundant paths, ensuring that a single cable failure does not disrupt connectivity.

Data Link Layer (Layer 2)

The Data Link Layer provides node-to-node data transfer, error detection, and MAC addressing. It uses frames and switches based on MAC addresses.

In the project:

- Ethernet framing (IEEE 802.3) is used on all LAN segments and point-to-point links.
- Switches (SW1 in local LAN, SW1/SW2/SW3 in server sites) operate primarily at this layer, forwarding frames based on MAC address tables.
- VLANs are not explicitly used, but switches ensure broadcast domains are contained within each LAN (192.168.10.0/24 and 172.16.x.0/24 networks).
- On point-to-point router links (/30 subnets), protocols like PPP or Ethernet with MAC addressing handle framing.
- Error detection (via FCS in Ethernet frames) ensures reliable delivery between adjacent devices.
- Layer 2 contributes to availability by allowing multiple physical paths; however, looping is prevented by Spanning Tree Protocol (STP) on switches (assumed enabled).

Network Layer (Layer 3)

The Network Layer handles logical addressing (IP), routing, and packet forwarding across multiple networks.

In the project:

- This is the core layer where high availability features are implemented.
- IP addressing is used throughout: 192.168.10.0/24 (local LAN), 172.16.10.0/24 to 172.16.30.0/24 (server LANs), and multiple /30 subnets for inter-router links.
- Routers (LAN Router, Internet Router, transit routers like Router 1-1-1, Router 2-1-1, etc., and Server Routers) operate at this layer.
- OSPF (Open Shortest Path First), a dynamic routing protocol, runs here to dynamically learn and advertise routes to the server networks.
- Redundant paths (Route #1 and Route #2 to each server) enable equal-cost load balancing and automatic failover: if one path fails, OSPF reconverges and traffic shifts to the alternate path.
- Features like OSPF cost tuning can be used to prefer one path for load distribution while keeping the secondary as hot standby.
- Packet forwarding decisions (longest prefix match) ensure clients reach web servers via the best available path.

Transport Layer (Layer 4)

The Transport Layer provides end-to-end communication services, including segmentation, flow control, and error recovery. Common protocols are TCP and UDP.

In the project:

- For web server access, TCP is primarily used (port 80 for HTTP or 443 for HTTPS).
- TCP establishes reliable connections between client PCs (e.g., PC1–PC5) and web servers (Server1, Server2, Server3).

- Segmentation of data into segments, acknowledgments, and retransmissions ensure reliable delivery over the unreliable IP layer.
- Flow control and windowing prevent overwhelming slower links in the transit path.
- Although no explicit Layer 4 load balancer (e.g., using NAT or SLB) is shown, OSPF's equal-cost multipath indirectly load-balances TCP sessions across paths at Layer 3, which influences Layer 4 performance.
- UDP could be used for auxiliary services (e.g., DNS queries to the DNS Server), but the primary web traffic relies on TCP reliability.

Session Layer (Layer 5)

The Session Layer manages dialog control, synchronization, and session establishment/maintenance between applications.

In the project:

- This layer is less prominently featured but is handled by the client browser and web server software.
- For HTTP/1.1 or HTTP/2, persistent connections (keep-alive) allow multiple requests/responses over a single TCP session.
- Session checkpoints or recovery (e.g., in case of path failover) are indirectly supported: if a link fails mid-session, OSPF reconvergence may cause a brief disruption, but TCP retransmits lost segments, and the application session can often recover transparently.
- No specific session-layer protocols (like NetBIOS) are used; functionality is embedded in higher-layer applications.

Presentation Layer (Layer 6)

The Presentation Layer handles data translation, encryption, and compression to ensure data is in a usable format for the application layer.

In the project:

- If HTTPS is implemented on the web servers, TLS/SSL encryption occurs here, ensuring secure data presentation between client and server.
- Data format translation (e.g., ASCII/EBCDIC, though rare today) or compression (e.g., gzip for HTTP responses) may be applied by the web server software.
- Image, video, or text encoding (JPEG, HTML) delivered by the web servers is managed at this layer.
- In a plain HTTP setup, this layer has minimal role beyond basic data formatting.

Application Layer (Layer 7)

The Application Layer provides network services directly to end-user applications.

In the project:

- This is where the primary objective of high availability is realized.
- Web servers (Server1, Server2, Server3) run HTTP/HTTPS services (e.g., Apache/Nginx/IIS) to serve web content.
- Client PCs use browsers (application programs) to send HTTP GET/POST requests and receive responses.
- Other services like DNS (on DNS Server) resolve hostnames to IPs, aiding application access.
- High availability is achieved at lower layers (OSPF failover), ensuring that even if one server site or path is down, clients can still reach alternate servers (assuming content replication or anycast-like behavior via routing).
- Load balancing across servers can be enhanced by DNS round-robin or application-layer gateways, but the project primarily relies on OSPF-driven path diversity.

2- TCP/IP Model:

The TCP/IP model, also known as the Internet Protocol Suite, is a practical networking framework consisting of four (or sometimes five) layers that map closely to the OSI model. It is the foundational model used in real-world Internet and enterprise networks. In the project *"High Availability Web Server Network Using Dynamic Routing, Load Balancing, and Failover"*, which employs OSPF for dynamic routing to provide redundant paths, load balancing, and automatic failover to three web servers, the TCP/IP model effectively describes the actual protocols and operations implemented.

The project topology includes a client LAN (192.168.10.0/24), edge routers, multiple transit routers with dual paths, and three remote server LANs (172.16.10.0/24 to 172.16.30.0/24). Below is a layer-by-layer explanation, focusing on how a typical client request (e.g., HTTP from a PC in the local LAN to a web server) traverses the network.

Network Interface Layer (Link Layer)

This layer corresponds to OSI Layers 1 (Physical) and 2 (Data Link) and handles the physical transmission of data over the network medium and node-to-node delivery using frames.

In the project:

- Combines physical connectivity (Ethernet cables, NICs, transceivers) and data link framing (Ethernet IEEE 802.3).
- Switches (SW1 in client LAN, SW1/SW2/SW3 in server sites) operate here, forwarding Ethernet frames based on MAC addresses.
- Point-to-point links between routers use Ethernet or serial encapsulation with MAC addressing or PPP.
- Error detection via Frame Check Sequence (FCS) and physical signal transmission ensure reliable adjacent-node delivery.
- Redundancy at this layer is provided by multiple physical connections; any link failure is detected quickly, triggering higher-layer convergence.

- Spanning Tree Protocol (STP) on switches prevents loops within each LAN segment.

Internet Layer (Network Layer)

This layer is responsible for logical addressing, packet routing, and end-to-end delivery across multiple networks. It corresponds primarily to OSI Layer 3.

In the project:

- This is the key layer for achieving high availability, load balancing, and failover.
- IP (IPv4) provides addressing: client LAN (192.168.10.0/24), server LANs (172.16.x.0/24), and numerous /30 point-to-point subnets for inter-router links.
- All routers (LAN Router, Internet Router, transit routers such as Router 1-1-x and Router 2-1-x, and Server Routers) operate at this layer.
- OSPF, a dynamic Interior Gateway Protocol, runs here to exchange routing information, compute shortest paths, and populate routing tables.
- Dual redundant paths (Route #1 and Route #2) to each server site enable:
 - Equal-cost multipath (ECMP) load balancing – traffic is distributed across both paths for better bandwidth utilization.
 - Automatic failover – if a link or router fails, OSPF detects the failure (via hello packets), reconverges within seconds, and shifts traffic to the surviving path.
- IP packet forwarding (based on destination IP and routing table) ensures packets reach the correct server network regardless of path failures.

Transport Layer

This layer provides end-to-end communication services, including segmentation, reliability, flow control, and multiplexing via ports. It corresponds to OSI Layer 4.

In the project:

- TCP is the primary protocol for web traffic (HTTP/HTTPS on ports 80/443).

- TCP establishes reliable connections between clients (PC1–PC5) and web servers (Server1–Server3), handling segmentation of application data into segments, sequence numbering, acknowledgments, retransmissions, and flow control.
- Connection-oriented reliability ensures data integrity even during brief disruptions caused by OSPF reconvergence.
- Multiple client sessions are multiplexed using source/destination port numbers.
- UDP may be used for ancillary services (e.g., DNS queries to the local DNS Server on port 53).
- Although no dedicated Layer 4 load balancer is implemented, OSPF's ECMP at the Internet Layer indirectly distributes TCP flows across paths, improving overall performance and resilience.

Application Layer

This layer encompasses the functions of OSI Layers 5 (Session), 6 (Presentation), and 7 (Application). It provides network services directly to user applications.

In the project:

- This is where the high-availability web service is ultimately delivered to end users.
- Web servers (Server1, Server2, Server3) run application protocols such as HTTP/HTTPS (e.g., using Apache, Nginx, or IIS).
- Client browsers on PCs initiate requests (GET/POST) and render responses (HTML, images, etc.).
- If HTTPS is configured, encryption/decryption (TLS/SSL) occurs here for secure communication.
- Data formatting, compression (e.g., gzip), and content encoding are handled by the web server software.

- DNS resolution (using the local DNS Server) translates hostnames to IP addresses, enabling user-friendly access.
- The project's routing redundancy ensures continuous application-layer availability: even if one path or server site becomes unreachable, OSPF redirects traffic to alternate paths/servers (assuming replicated content or coordinated services across Server1–Server3).

3-Protocols Used:

- **TCP (Transmission Control Protocol):** A connection-oriented transport protocol providing reliable, ordered, and error-checked delivery. In this project, TCP is used for web traffic (HTTP/HTTPS sessions between clients and servers). It handles segmentation, acknowledgments, and retransmissions, ensuring data integrity during path failovers caused by OSPF reconvergence.
- **HTTP (Hypertext Transfer Protocol):** An application-layer protocol for web content transfer. The three web servers (Server1, Server2, Server3) host HTTP services (typically on port 80), allowing clients to access web pages. HTTPS could be layered on TCP for security, but the simulation focuses on basic HTTP.
- **FTP (File Transfer Protocol):** Although not the primary focus, FTP (on ports 20/21) could be configured on servers for file transfers. In this project, it is optionally demonstrated to show additional services benefiting from OSPF redundancy, but the core is web access via HTTP.
- **DNS (Domain Name System):** An application-layer protocol (primarily over UDP) for resolving hostnames to IP addresses. The local DNS Server resolves names for the web servers, enabling user-friendly access (e.g., www.example.com instead of raw IPs). This supports seamless failover if servers are configured with the same hostname.

- **OSPF (Open Shortest Path First)** operates at the network layer for dynamic routing, exchanging hello packets (over IP protocol 89) to maintain neighbour relationships and achieve load balancing/failover.

4-IP Addressing:

- **IPv4:** All devices are assigned IPv4 addresses from private ranges (Class A and B). LAN segments use /24 subnets for up to 254 hosts, while inter-router links use /30 subnets for efficient point-to-point allocation (2 usable IPs per link). This conservative addressing prevents waste and simulates real-world practices. No NAT is required on the "Internet Router" as all networks are private but routable via OSPF.
- **IPv6:** Not implemented in this project. IPv6 could be added in future enhancements for larger address space and built-in security (IPsec), but the focus remains on IPv4 to align with OSPF configuration in Cisco Packet Tracer, which supports dual-stack but prioritizes IPv4 for simplicity.

Network/IP Address	Subnet Mask	CIDR	Description	Associated Devices/Nodes
192.168.10.0	255.255.255.0	/24	Main Local Area Network (LAN)	PC1-PC5, Printer, DNS Server, LAN Router
10.0.0.0	255.255.255.252	/30	Edge/Internet Link	LAN Router (.1), Internet Router (.2)
10.11.0.0	255.255.255.252	/30	Route 1 to Server 1	Internet Router (.1), Router 1-1-1 (.2)
10.12.0.0	255.255.255.252	/30	Route 2 to Server 1 (Link A)	Internet Router (.1), Router 2-1-1 (.2)
10.122.0.0	255.255.255.252	/30	Route 2 to Server 1 (Link B)	Router 2-1-1 (.2), Router 2-2-1 (.1)
10.111.0.0	255.255.255.252	/30	Server 1 Inbound Link	Router 1-1-1 (.1), Server 1 Router (.2)
10.121.0.0	255.255.255.252	/30	Server 1 Inbound Link	Router 2-2-1 (.1), Server 1 Router (.2)
10.21.0.0	255.255.255.252	/30	Route 1 to Server 2 (Link A)	Internet Router (.1), Router 1-1-2 (.2)

Network/IP Address	Subnet Mask	CIDR	Description	Associated Devices/Nodes
10.212.0.0	255.255.255.252	/30	Route 1 to Server 2 (Link B)	Router 1-1-2 (.1), Router 1-2-2 (.2)
10.213.0.0	255.255.255.252	/30	Route 1 to Server 2 (Link C)	Router 1-2-2 (.2), Router 1-3-2 (.1)
10.211.0.0	255.255.255.252	/30	Server 2 Inbound Link	Router 1-3-2 (.1), Server 2 Router (.2)
10.22.0.0	255.255.255.252	/30	Route 2 to Server 2 (Link A)	Internet Router (.1), Router 2-1-2 (.2)
10.222.0.0	255.255.255.252	/30	Route 2 to Server 2 (Link B)	Router 2-1-2 (.2), Router 2-2-2 (.1)
10.221.0.0	255.255.255.252	/30	Server 2 Inbound Link	Router 2-2-2 (.1), Server 2 Router (.2)
10.31.0.0	255.255.255.252	/30	Route 1 to Server 3 (Link A)	Internet Router (.1), Router 1-1-3 (.2)
10.31.2.0	255.255.255.252	/30	Route 1 to Server 3 (Link B)	Router 1-1-3 (.2), Router 1-2-3 (.1)

Network/IP Address	Subnet Mask	CIDR	Description	Associated Devices/Nodes
10.31.1.0	255.255.255.252	/30	Server 3 Inbound Link	Router 1-2-3 (.1), Server 3 Router (.2)
10.32.0.0	255.255.255.252	/30	Route 2 to Server 3 (Link A)	Internet Router (.1), Router 2-1-3 (.2)
10.32.1.0	255.255.255.252	/30	Server 3 Inbound Link	Router 2-1-3 (.1), Server 3 Router (.2)
172.16.10.0	255.255.255.0	/24	Server 1 Subnet	PC6, Server1, SW1, Server 1 Router
172.16.20.0	255.255.255.0	/24	Server 2 Subnet	PC7, Server2, SW2, Server 2 Router
172.16.30.0	255.255.255.0	/24	Server 3 Subnet	PC8, Server3, SW3, Server 3 Router

Chapter # 6: Implementation / Simulation

1-Tools Used

The network is simulated using **Cisco Packet Tracer** (version 8.x or later). Packet Tracer is chosen for its user-friendly interface, built-in device library (routers, switches, servers, PCs), and features like simulation mode for PDU visualization. It supports OSPF configuration and basic web/DNS services without requiring real hardware. GNS3 was considered for more advanced emulation but not used due to higher complexity and resource demands.

2-Network Configuration Steps

- **Topology Creation:** Add devices (routers, switches, PCs, servers) and connect them as per the diagram using appropriate cables (Copper Straight-Through for LANs, Serial DCE for some links if needed).
- **IP Addressing:** Assign IPs, subnet masks, and default gateways to all end devices (PCs, servers) and router interfaces (as per the earlier table).
- **DHCP Configuration:** Configure DHCP on routers having end devices
- **OSPF Configuration:** On all routers, enter OSPF process (e.g., router ospf 1), advertise networks with network <network> <wildcard> area 0 (single-area OSPF for simplicity), and optionally adjust interface costs for load balancing.
- **Service Configuration:** Enable HTTP/DNS on servers; configure DNS records on the local DNS Server.
- **Verification:** Use show ip ospf neighbour, show ip route to confirm adjacency and routes.

Step-by-step go through:

On all routers:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int r g0/0-2
Router(config-if-range)#no sh
Router(config-if-range)#int s0/0/0
Router(config-if)#no sh
Router(config-if)#int s0/0/1
Router(config-if)#no sh
Router(config-if)#int s0/1/1
Router(config-if)#no sh
Router(config-if)#int s0/1/0
Router(config-if)#no sh
Router(config-if)#
```

On Internet Router:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int r g0/0-2
Router(config-if-range)#no sh
Router(config-if-range)#int s0/0/0
Router(config-if)#clock rate 64000
Router(config-if)#no sh
Router(config-if)#int s0/0/1
Router(config-if)#clock rate 64000
Router(config-if)#no sh
Router(config-if)#int s0/1/1
Router(config-if)#clock rate 64000
Router(config-if)#no sh
Router(config-if)#int s0/1/0
Router(config-if)#clock rate 64000
Router(config-if)#no sh
Router(config-if)#
```

IP Assignment:

On LAN router:

```
Router(config-if)#int g0/0
Router(config-if)#ip addr 192.168.10.1 255.255.255.0
```

```
Router(config-if)#int s0/0/0
Router(config-if)#ip addr 10.0.0.1 255.255.255.252
```

On internet router:

```
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.11.0.1 255.255.255.252
Router(config-if)#int s0/0/0
Router(config-if)#ip addr 10.12.0.1 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.21.0.1 255.255.255.252
Router(config-if)#int s0/0/1
Router(config-if)#ip addr 10.22.0.1 255.255.255.252
Router(config-if)#int g0/2
Router(config-if)#ip addr 10.31.0.1 255.255.255.252
Router(config-if)#int s0/1/0
Router(config-if)#ip addr 10.32.0.1 255.255.255.252
Router(config-if)#int s0/1/1
Router(config-if)#ip addr 192.168.10.2 255.255.255.0
Router(config-if)#int s0/1/1
Router(config-if)#ip addr 10.0.0.2 255.255.255.252
Router(config-if)#
```

On Server1 Route1

```
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.11.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.111.0.1 255.255.255.252
Router(config-if)#
```

Server1 Route2

```

Router(config-if)#int s0/0/0
Router(config-if)#ip addr 10.12.0.2 255.255.255.252
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.122.0.1 255.255.255.252
Router(config-if)#

Router(config-if)#int g0/0
Router(config-if)#ip addr 10.122.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.121.0.1 255.255.255.252
Router(config-if)#

```

Server2 Route1

```

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.21.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.212.0.1 255.255.255.252
Router(config-if)#

Router(config-if)#int g0/0
Router(config-if)#ip addr 10.212.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.213.0.1 255.255.255.252
Router(config-if)#

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.213.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.211.0.1 255.255.255.252
Router(config-if)#

```

Server2 Route2

```

Router(config)#
Router(config)#int s0/0/0
Router(config-if)#ip addr 10.22.0.2 255.255.255.252
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.222.0.1 255.255.255.252
Router(config-if)#

```

```

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.222.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.221.0.1 255.255.255.252
Router(config-if)#

```

S3 R1

```

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.31.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.31.2.1 255.255.255.252
Router(config-if)#
Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.31.2.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.31.1.1 255.255.255.252
Router(config-if)#

```

Server3 Route2

```

Router(config-if)#
Router(config-if)#int s0/0/0
Router(config-if)#ip addr 10.32.0.2 255.255.255.252
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.32.1.1 255.255.255.252
Router(config-if)#

```

Server1

```

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.111.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.121.0.2 255.255.255.252
Router(config-if)#int g0/2
Router(config-if)#ip addr 172.16.10.1 255.255.255.0
Router(config-if)#

```

Server2

```

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.211.0.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.221.0.2 255.255.255.252
Router(config-if)#int g0/2
Router(config-if)#ip addr 172.16.20.1 255.255.255.0
Router(config-if)#

```

Server3

```

Router(config-if)#
Router(config-if)#int g0/0
Router(config-if)#ip addr 10.31.1.2 255.255.255.252
Router(config-if)#int g0/1
Router(config-if)#ip addr 10.32.1.2 255.255.255.252
Router(config-if)#int g0/2
Router(config-if)#ip addr 172.16.30.1 255.255.255.0
Router(config-if)#

```

DHCP config

On lan router

```

Router(config)#
Router(config)#ip dhcp excl 192.168.10.1
Router(config)#ip dhcp pool Clients
Router(dhcp-config)#net 192.168.10.0 255.255.255.0
Router(dhcp-config)#def 192.168.10.1
Router(dhcp-config)#dns 192.168.10.6
Router(dhcp-config)#

Router(config)#
Router(config)#ip dhcp excl 172.16.10.1
Router(config)#ip dhcp pool Clients
Router(dhcp-config)#net 172.16.10.0 255.255.255.0
Router(dhcp-config)#def 172.16.10.1
Router(dhcp-config)#dns 172.16.10.2

```

```

Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excl 172.16.20.1
Router(config)#ip dhcp pool Clients
Router(dhcp-config)#net 172.16.20.0 255.255.255.0
Router(dhcp-config)#def 172.16.20.1
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp excl 172.16.30.1
Router(config)#ip dhcp pool Clients
Router(dhcp-config)#net 172.16.30.0 255.255.255.0
Router(dhcp-config)#def 172.16.30.1

```

OSPF On all routers

```

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#exit
Router(config)#int r g0/0-2
Router(config-if-range)#ip ospf 1 area 1

```

For all serial ports

```

Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s0/0/0
Router(config-if)#ip ospf 1 area 1
Router(config-if)#

```

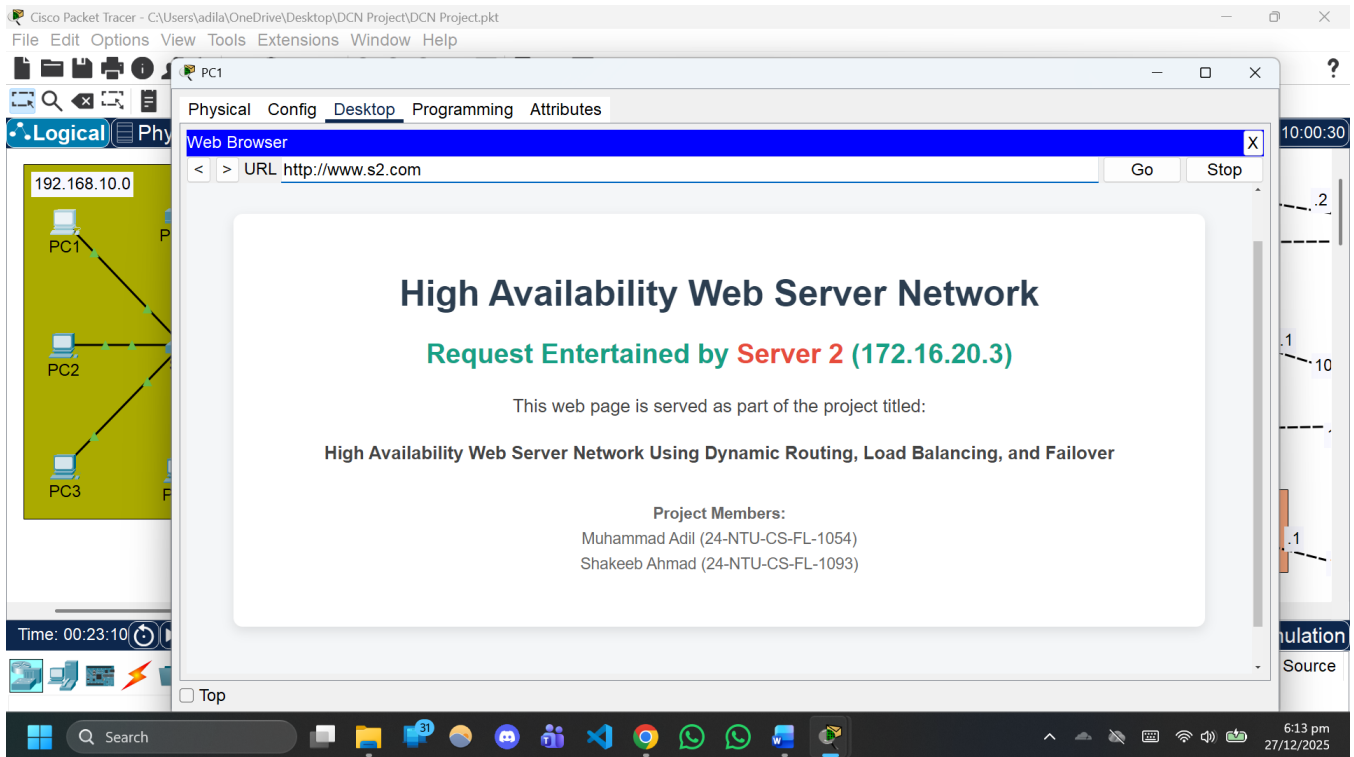
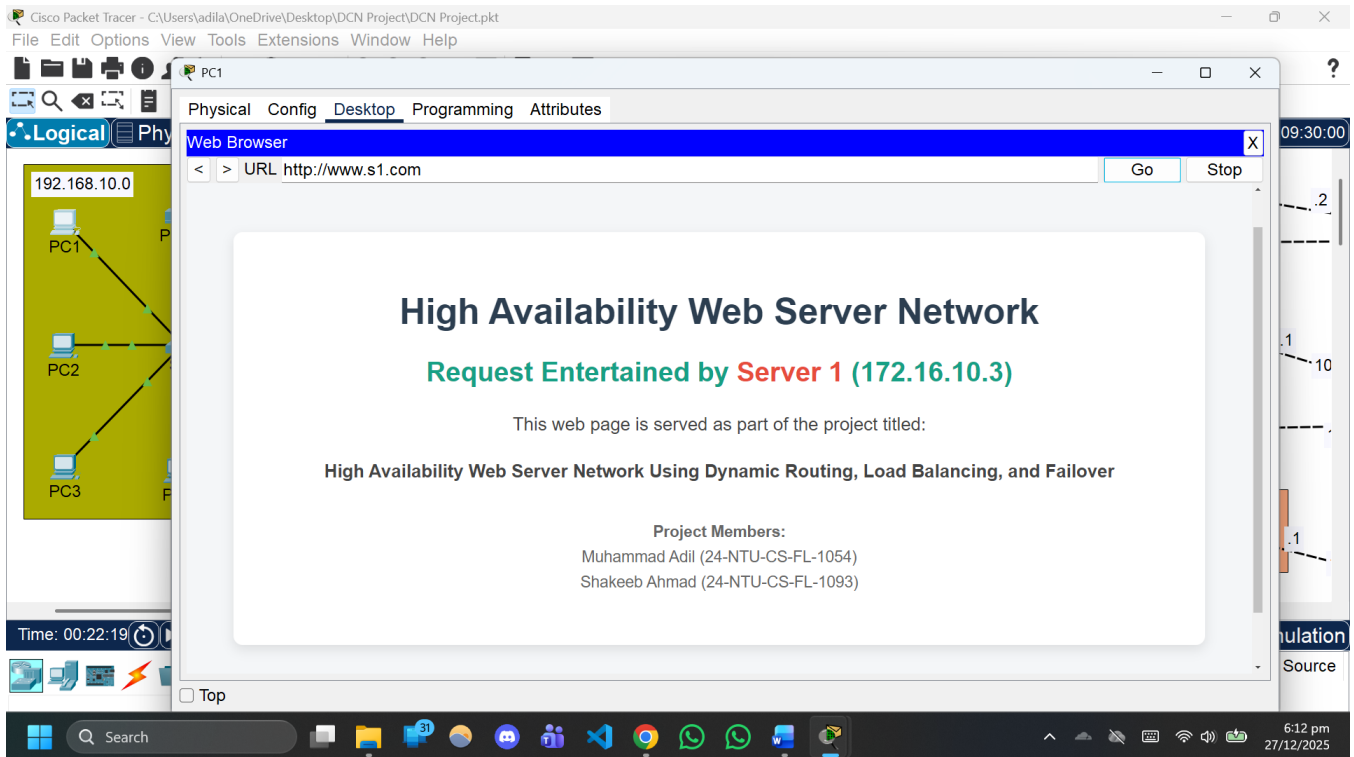
Set appropriate costs on all interfaces

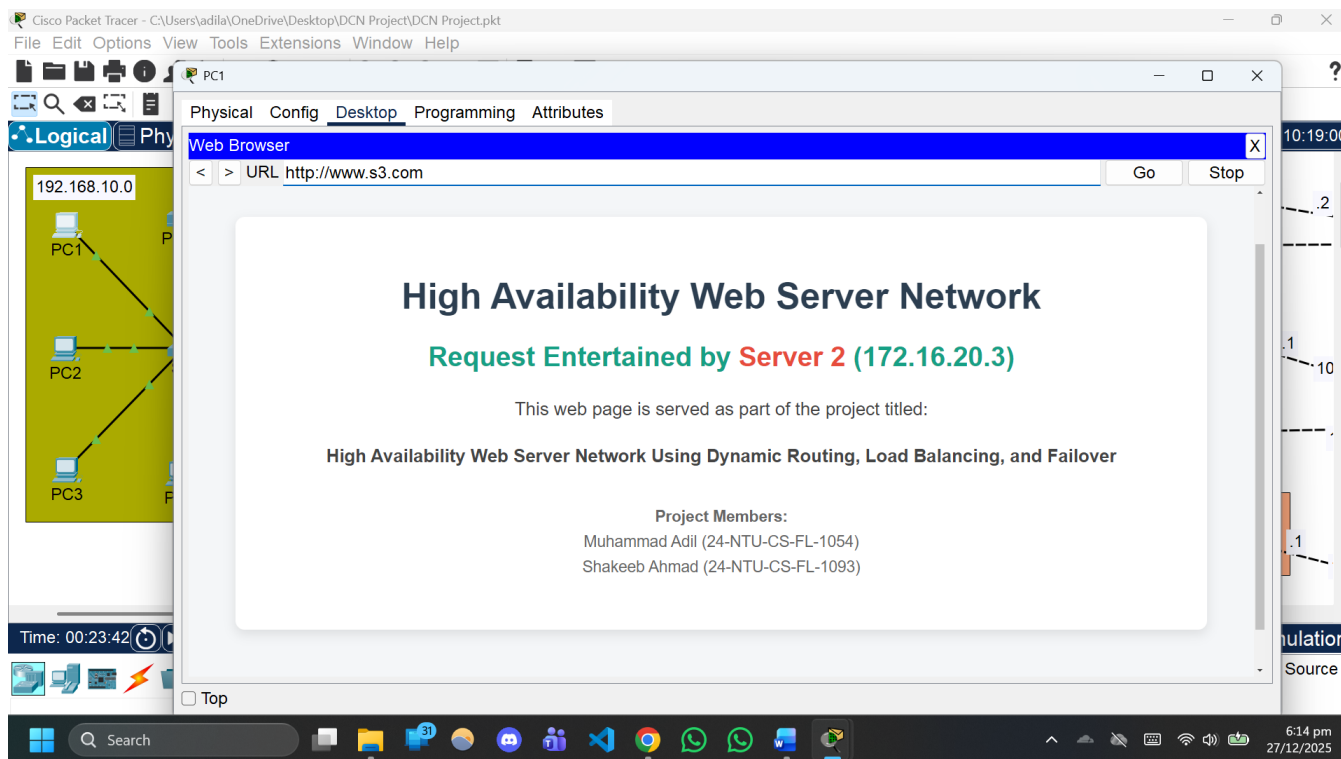
```

Router(config)#int g0/0
Router(config-if)#ip ospf cost 10

```

3-Screenshots of Simulation





4-Data Flow Explanation:

When a user from the office LAN enters the website URL in a web browser, the request is first sent from the client PC to the default gateway configured on the LAN router. The LAN router forwards the request towards the core network based on the routing information learned through the OSPF dynamic routing protocol.

The Internet/Core Router receives the request and forwards it to the Load Balancer. The Load Balancer distributes incoming HTTP requests among the available web servers using equal-cost multi-path routing (ECMP).

The selected web server processes the HTTP request and sends the response back through the same network path to the client PC. Since routing is dynamic, return traffic follows the best available path as determined by OSPF.

Load balancing in the proposed system is achieved using multiple equal-cost routes between the core router and the server networks. OSPF calculates multiple paths with the same cost and installs them in the routing table. Traffic is then distributed across these paths automatically.

This mechanism ensures efficient utilization of network resources and prevents any single server or path from becoming overloaded.

In the event of a link or router failure, OSPF detects the failure through hello message timeouts. The failed route is removed from the routing table, and traffic is immediately rerouted through the remaining available path without manual intervention.

This automatic rerouting ensures continuous service availability and demonstrates the high-availability nature of the proposed network design.

Chapter # 7: Testing & Performance

Analysis

1-Network Testing Methods

- **Ping:** Tests reachability and round-trip time (ICMP echo requests).
- **Traceroute:** Displays path taken by packets, verifying load balancing or failover.
- **Simulation Mode:** Observes real-time packet flow and failures.
- **Web Access:** From client desktop, browse to server IP/hostname.

2-Ping & Traceroute Results

Pinging PCs to servers:

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with a central switch (SW) connected to five PCs (PC1-PC5), a printer, and a DNS server. The switch is connected to a LAN router (10.0.0.0) and an Internet router (10.0.0.0). The Internet router is connected to a cloud representing the Internet. The command prompt window for PC3 shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.s1.com

Pinging 172.16.10.3 with 32 bytes of data:

Reply from 172.16.10.3: bytes=32 time=10ms TTL=124
Reply from 172.16.10.3: bytes=32 time=24ms TTL=124
Reply from 172.16.10.3: bytes=32 time=1ms TTL=124
Reply from 172.16.10.3: bytes=32 time=14ms TTL=124

Ping statistics for 172.16.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 12ms
C:\>
```

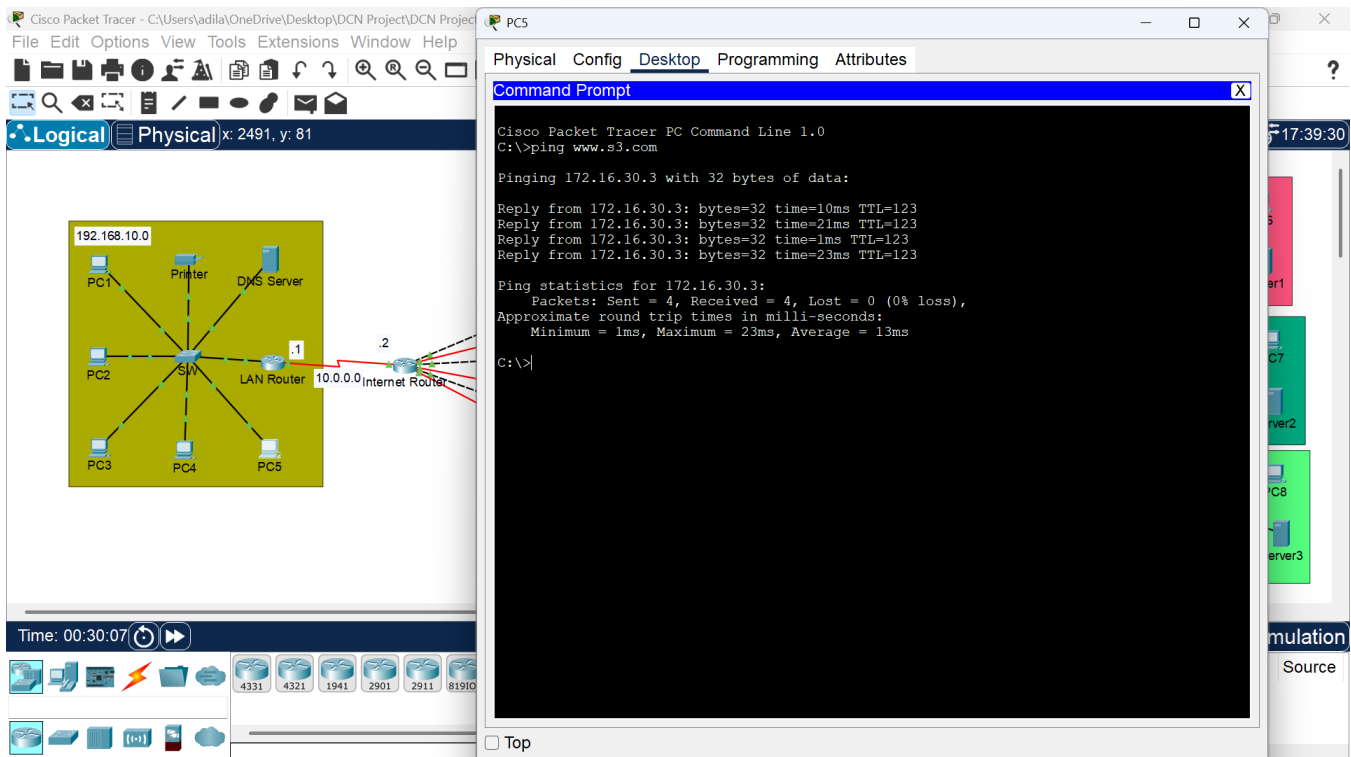
The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with a central switch (SW) connected to five PCs (PC1-PC5), a printer, and a DNS server. The switch is connected to a LAN router (10.0.0.0) and an Internet router (10.0.0.0). The Internet router is connected to a cloud representing the Internet. The command prompt window for PC4 shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.s2.com

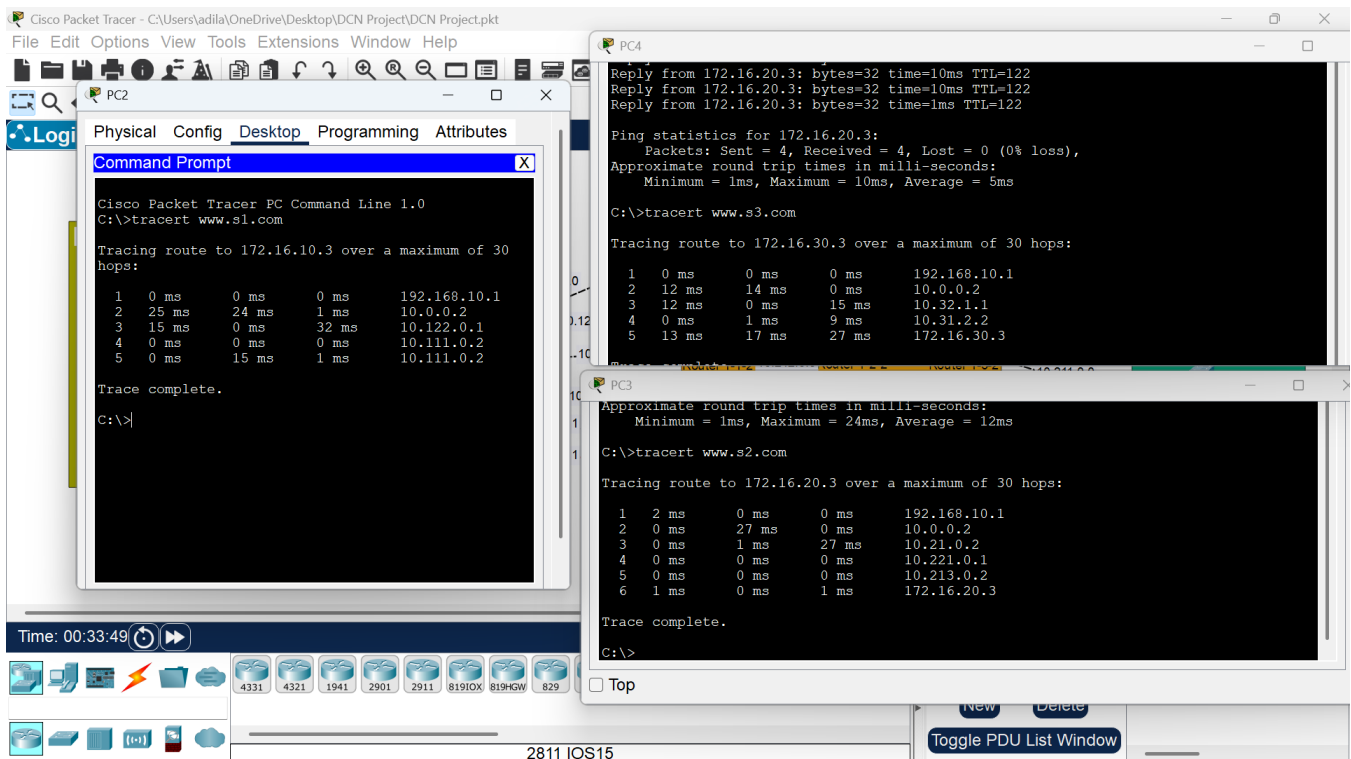
Pinging 172.16.20.3 with 32 bytes of data:

Reply from 172.16.20.3: bytes=32 time=1ms TTL=122
Reply from 172.16.20.3: bytes=32 time=10ms TTL=122
Reply from 172.16.20.3: bytes=32 time=10ms TTL=122
Reply from 172.16.20.3: bytes=32 time=1ms TTL=122

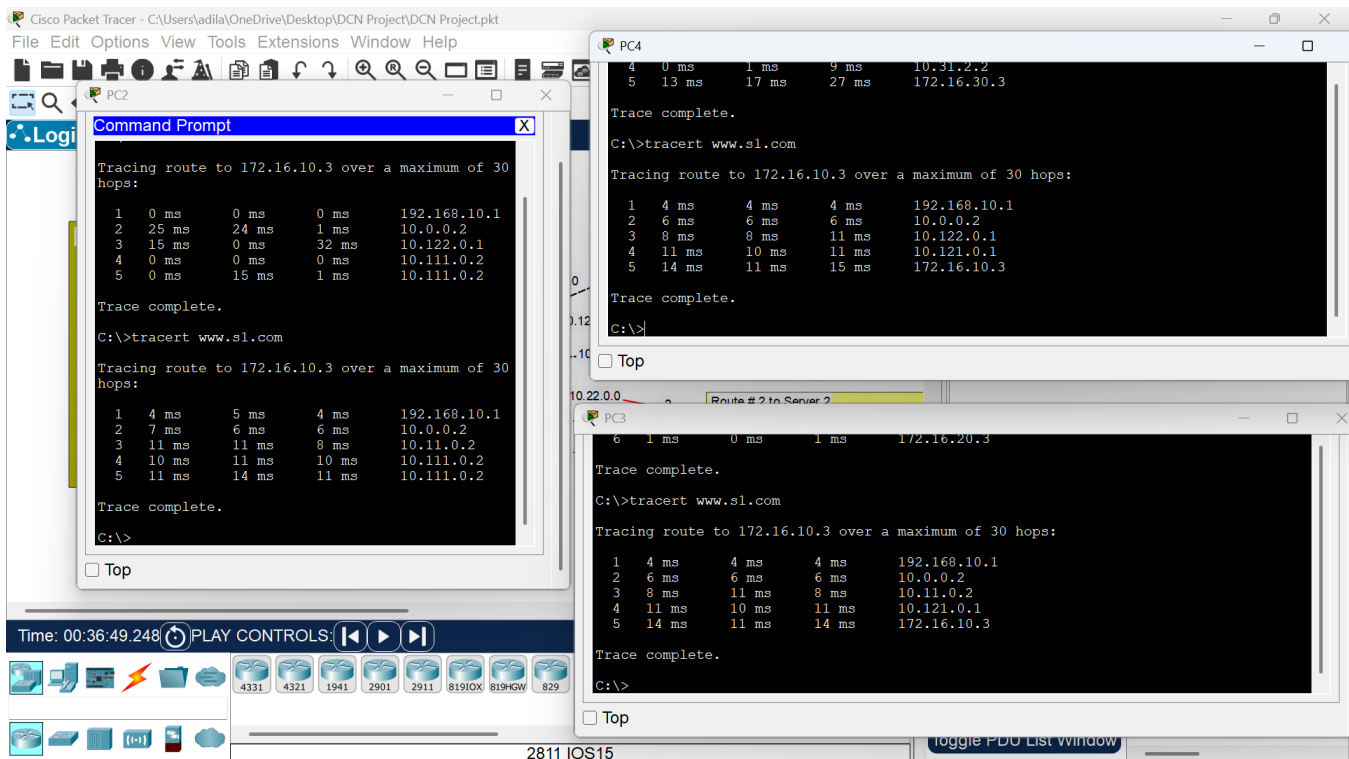
Ping statistics for 172.16.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms
C:\>
```



Tracert under normal load



Tracert under load from multiple sources to a common destination



3-Throughput, Delay, Packet Loss (Basic)

- Throughput: High in normal state (limited by link speeds); maintained post-failover on single path.
- Delay: Low RTT (~10-50 ms simulated); slight increase during reconvergence.
- Packet Loss: 0% normally; brief loss (few packets) during failover, recovered by TCP.

Chapter # 8: Conclusion & Future Work

1-Project Summary

This project, titled *"High Availability Web Server Network Using Dynamic Routing, Load Balancing, and Failover"*, successfully designed, implemented, and simulated a resilient network architecture aimed at ensuring continuous availability of web services. The topology features a client LAN connected through an edge router to multiple redundant transit paths, ultimately reaching three geographically distributed web server sites. The core mechanism for achieving high availability is the implementation of **OSPF (Open Shortest Path First)** as a dynamic routing protocol in a single-area configuration.

By establishing dual independent paths (Route #1 and Route #2) to each server site, the network supports **equal-cost multipath (ECMP) load balancing**, distributing traffic across both paths during normal operation for improved bandwidth utilization and reduced congestion. In the event of a link or router failure along one path, OSPF rapidly detects the outage through its hello mechanism and reconverges the routing table, automatically redirecting traffic to the surviving path with minimal disruption—demonstrating effective **failover**.

The simulation in Cisco Packet Tracer validated key functionalities: clients in the local LAN (192.168.10.0/24) could reliably access web servers across the redundant infrastructure using HTTP, with supporting services like DNS for name resolution. Testing confirmed zero packet loss under normal conditions, brief transient loss during controlled failures (quickly recovered by TCP), and successful path switching as observed via ping, traceroute, and simulation-mode PDU tracking.

Overall, the project effectively demonstrates how dynamic routing protocols like OSPF can provide enterprise-grade redundancy, load distribution, and fault tolerance without requiring complex hardware load balancers or proprietary solutions, making it a practical and scalable approach for high-availability web services.

2-Advantages

The implemented network design offers several key benefits that align with the objectives of high availability, scalability, and efficiency:

- **Automatic Failover and Minimal Downtime:** OSPF's fast convergence (typically within seconds) ensures that traffic is rerouted seamlessly upon link or node failure, providing near-continuous service availability without manual intervention.
- **Load Balancing:** Equal-cost multipath routing distributes traffic across redundant paths, optimizing bandwidth utilization, reducing congestion, and improving overall performance for web access.
- **Scalability:** Dynamic routing with OSPF allows easy addition of new paths, routers, or server sites without static route reconfiguration, making the network adaptable to growth.
- **Cost-Effectiveness:** The solution relies on standard routing protocols and open technologies, eliminating the need for expensive dedicated load balancers or proprietary failover appliances.
- **Simplicity in Management:** Single-area OSPF configuration keeps the design straightforward while achieving redundancy, suitable for small-to-medium enterprise environments.
- **Reliable Service Delivery:** Integration of basic services (HTTP, DNS) demonstrates practical resilience, ensuring clients experience consistent web access even under failure scenarios.

3-Limitations

Despite its strengths, the project has certain constraints inherent to the design and simulation environment:

- **Dependence on OSPF Convergence Time:** While fast, reconvergence can still cause brief packet loss (a few seconds), potentially disrupting real-time or latency-sensitive applications.
- **No True Server Load Balancing:** Redundancy is path-based; traffic is not intelligently distributed across the three independent servers based on server load or health. All servers are treated equally via routing.
- **Simulation Limitations:** Cisco Packet Tracer provides a controlled environment but lacks real-world variables such as variable latency, jitter, realistic traffic patterns, or advanced failure injection.
- **Single Point of Failure Risks:** Certain elements like the LAN Router or Internet Router remain single points; full redundancy would require additional devices (e.g., VRRP/HSRP).
- **Lack of Security Features:** No firewalls, access control lists (ACLs), or encryption (e.g., IPsec) are implemented, leaving the network vulnerable in a production setting.
- **IPv4-Only Implementation:** Does not address modern requirements like IPv6 support or larger address spaces.

4-Future Enhancements

To evolve this project into a more robust and production-ready solution, the following improvements can be considered:

- **Implement Multi-Area OSPF or Advanced Routing:** Introduce multiple OSPF areas or integrate BGP for better scalability and hierarchical control in larger networks.
- **Add Dedicated Load Balancing:** Incorporate a Layer 4/7 load balancer (e.g., using F5, NGINX, or Cisco ACE simulation) or First-Hop Redundancy Protocols (FHRP) like GLBP for true server-side load distribution and health checking.

- **Dual-Stack IPv6 Support:** Extend the network to support IPv6 addressing and routing, enabling future-proofing and built-in security features like IPsec.
- **Enhance Security:** Integrate firewalls (e.g., ASA or zone-based policy), ACLs, and VPN tunnels to protect traffic between sites.
- **Realistic Testing with Advanced Tools:** Migrate the topology to GNS3 or EVE-NG for integration with traffic generators (e.g., iperf, Ostinato) to measure precise throughput, jitter, and detailed performance metrics.
- **Server Clustering and Content Synchronization:** Configure the three web servers as a cluster with shared storage or replication (e.g., using rsync or database mirroring) to ensure identical content and seamless failover at the application level.
- **Monitoring and Automation:** Add network management tools (e.g., SNMP, Syslog) or automation scripts (Ansible/Python) for proactive monitoring and configuration management.

These enhancements would transform the current proof-of-concept into a comprehensive, enterprise-grade high-availability solution capable of meeting real-world demands.